Comparing Traditional and Data-Driven Batting Orders Using MLB Statistics

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Research Question

Does a data-driven batting order based on predicted player performance produce more expected runs than a traditional batting order?

Application

- Many data analyst who work in Baseball analytics say that the “traditional” baseball batting lineup does not result in the most runs per game
- Many experts state that having the best to worst batters in chronological order results in the most runs

Data

- There is publicly available baseball player data on Baseball Reference
- I will use AVG, OBP, SLG, HR, H, BB, 2B, 3B, RBI, SB, R from 2023-2026 as my data
- Most of this data is publicly available and involves minimal cleaning
- Some limitations would be high correlation between these variables

Data

- I will choose the 12 best hitters from 5 specific teams
- I will choose two good offensive, two bad, and one neutral MLB team to also see the difference on how this model affects good vs bad teams
- Teams: Dodgers, Cubs, Cardinals, Angles, Giants

Proposed Approach

- Predictive Modeling
- Use PCA to reduce correlated offensive statistics into a smaller number of components
- Capture different aspects of player performance

Create two batting-order strategies:

- Traditional batting order
- Data-driven batting order

Monte Carlo simulation to estimate expected runs under each strategy

Anticipated pitfalls:

- Data leakage when predicting future performance
- Multicollinearity among offensive statistic

Scope Definition

- Scope would be to use PCA and Monte Carlo methods to figure out if a traditional lineup is better suited for MLB teams than a chronological lineup
- Some stretch goals would be to compare results across teams with different levels of offensive strength or test different definitions of a "traditional" batting order
- Some “Out of Scope” things would be to find the most optimal lineup and which specific lineup maximizes runs per game
- I believe that this project is manageable because the project compares two strategies rather than optimizing millions of possible lineups and it still allows room for the DS to expand the predictive model or simulation if time permits

Final deliverable & audience

- R Markdown
- I would like to see PCA results and player profiles
- Traditional vs. data-driven batting orders
- Monte Carlo simulation results
- Comparison of expected runs
- Audience: Baseball data analyst and MLB coaches
- Usefulness: Provide evidence on whether a “non-traditional” -based batting order could improve expected run production compared with a traditional lineup

How you'll manage the project

- I will have check-ins with my Data Scientist to ensure that the project is running smoothly
- The project uses publicly available baseball statistics and does not involve sensitive personal information.
- Check that the data is appropriate, the PCA is interpreted correctly, and review whether the simulation assumptions are reasonable
- If the DS encounters a roadblock we will work on the problem together and figure it out